

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Summer Supplementary Semester Examination 2024 Course: B. Tech. Semester: VII Branch : Computer Engineering/ Computer Science And Engineering Subject Code & Name: Deep Learning (BTCOE705B) Max Marks: 60 Date: 10/07/2024 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Distinguish between supervised and unsupervised learning approach.	Remember	6
B)	State and explain Bayes rule for classification.	Understand	6
C)	What are the different steps used in typical Deep Learning model.	Understand	6
Q.2	Solve Any Two of the following.		12
A)	What is neuron? What is neural network?	Understand	6
B)	Explain different optimization techniques with their advantages and disadvantages.	Remember	6
C)	What is back propagation learning?	Understand	6
Q. 3	Solve Any Two of the following.		12
A)	How weights are updated at output layer for multilayer neural network?	Apply	6
B)	What is Autoencoder? Explain Undercomplete Autoencoder in detail.	Understand	6
C)	State advantages of Convolutional Neural Network over multilayer perceptron.	Analyze	6
Q.4	Solve Any Two of the following.		12
A)	Explain the concept of layer by layer pretraining mechanism.	Remember	6
B)	What is vanishing gradient problem? What are the challenges in Gradient Descent?	Remember	6
C)	What is cross entropy loss?	Remember	6
Q. 5	Solve Any Two of the following.		12

A)	Why normalization is required? What is batch normalization?	Understand	6
B)	State some applications of machine learning and deep learning.	Remember	6
C)	Design neuron for AND and OR operation.	Understand	6
	*** End ***		